

Distilling Lorentz-Equivariant Transformers for Efficient Jet Tagging

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Abstract

We study whether the behavior of a Lorentz-equivariant L-GATr top-tagging teacher can be transferred to a compact non-equivariant transformer student. A 205k-parameter Particle Transformer-style student matches the teacher’s i.i.d. AUC within seed noise (0.9869 vs. 0.9870) while using about $5\times$ fewer parameters. Under Lorentz x -boosts of the test inputs, the distilled student remains more robust than a same-architecture scratch baseline, recovering roughly 40% of the boost-AUC gap between the scratch student and teacher. Controlled ablations over temperature, hard/soft mixing, hint losses, soft-target form and hand-engineered pairwise features show that the main gain comes from the student’s architectural bias rather than from a particular distillation recipe. These results suggest that distillation can reduce the computational cost of Lorentz-aware jet taggers while retaining part of their robustness under frame changes.

Keywords: knowledge distillation, jet tagging, Lorentz equivariance, top tagging, transformer

1 Introduction

Jet tagging classifies the parent particle that produced a collimated spray of hadronic decay products. Top tagging, the binary classification of hadronic top-quark jets against light-quark or gluon jets, is a standard benchmark for machine learning on collider point clouds [4]. Recent Lorentz-equivariant architectures are especially strong on this task: L-GATr [9] reaches AUC 0.9870 with about one million parameters by processing particle four-momenta as multivectors and producing a scalar classification logit.

The practical question is whether this symmetry-aware behavior can be made cheaper. L-GATr replaces ordinary transformer operations with multivector channels, geometric products and equivariant attention. These ingredients are useful, but they also make the model less convenient as a small surrogate or a fast ablation target. Meanwhile, the standard i.i.d. TopTagging leaderboard is nearly saturated: PELICAN [1], ParticleNet [6] and ParT [7] all approach L-GATr-level AUC without exact Lorentz equivariance. Matching the i.i.d. test set is therefore not enough to tell us whether the equivariant teacher has transferred anything beyond ordinary classification performance.

We ask whether knowledge distillation transfers any of the teacher’s Lorentz-robust behavior to a compact non-equivariant student. We train Particle Transformer-style and plain-transformer students from the L-GATr teacher, then evaluate them both on the standard i.i.d. test set and under deliberate Lorentz boosts of the test inputs. The teacher is treated primarily as a black box, accessible through its scalar logit; in one ablation we also use penultimate scalar invariants as a hint target. Our main contributions are:

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(1) Distillation closes the i.i.d. performance gap. A 205k-parameter Particle Transformer-style student matches the L-GATr teacher’s i.i.d. AUC within seed noise (0.9869 vs. 0.9870), while using about $5\times$ fewer parameters. This shows that the standard TopTagging decision boundary can be compressed into a much smaller non-equivariant student.

(2) Boost tests reveal partial behavioral transfer. Under Lorentz x -boosts of the test inputs, the distilled student is consistently more robust than a same-architecture scratch baseline. It recovers roughly 40% of the boost-AUC gap between the scratch student and the L-GATr teacher, indicating that distillation transfers more than i.i.d. classification accuracy.

(3) Student architecture matters more than the distillation recipe. Ablations over temperature, hard/soft mixing, hint losses, soft-target form and pairwise input features show that architecture is the main lever. The Particle Transformer-style pair bias improves performance more reliably than changing the supervision-side details.

Relation to prior work. Lorentz-equivariant jet taggers such as L-GATr [9], PELICAN [1] and LorentzNet [2] achieve strong TopTagging performance by building spacetime symmetry into the model, while ParticleNet [6] and ParT [7] show that non-equivariant architectures with strong geometric priors can approach the same i.i.d. accuracy. This makes TopTagging a useful but delicate setting: standard test AUC alone cannot tell whether a student has inherited symmetry behavior or has simply learned a saturated decision boundary.

The closest prior work is Liu et al. [5], who distill LorentzNet into DeepSet and MLP students and show that distillation can transfer Lorentz-invariance behavior even when the student is not equivariant by construction. Our work differs in the teacher, the student and the evaluation target. We use L-GATr, an equivariant transformer based on geometric algebra, as the teacher; distill into a compact Particle Transformer-style student with learned pairwise interaction biases; and evaluate both i.i.d. performance and explicit boost-robustness. We also ablate temperature, hard/soft mixing, hint losses, soft-target form and pairwise inputs to separate the effect of the distillation recipe from the architecture of the student.

2 Background

2.1 Lorentz-Equivariant Geometric Algebra Transformer

The teacher in our study is L-GATr [9], a Lorentz-equivariant geometric-algebra transformer. L-GATr embeds particle four-momenta as multivectors, augments them with reference/spurion tokens and processes them with equivariant linear maps, geometric products and geometric attention. Its classification head reads out a scalar logit, making it a natural source of Lorentz-aware supervision for a student that does not itself carry multivector representations.

2.2 Particle Transformer

The Particle Transformer (ParT) [7] is a strong non-equivariant baseline for jet tagging because it modifies ordinary self-attention with pairwise kinematic information. In addition to the usual query-key term, ParT adds a learned bias matrix derived from pair features such as angular distance, relative momentum and pair invariant mass:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + U\right) V.$$

Here U is produced from particle-pair features and gives the transformer a geometric prior without enforcing Lorentz equivariance. This makes a ParT-style model a useful student for our question: it is cheaper and non-equivariant, but not geometrically naive.

2.3 Knowledge Distillation

Knowledge distillation replaces, or mixes, the hard label objective with a soft target derived from the teacher [3]. Our default soft loss regresses the student logit toward the L-GATr logit, combined with ordinary supervised cross-entropy through a mixing weight α . We also test the standard temperature-softened KL objective, vary the temperature T , and add a FitNets-style hint loss [8] that aligns a student hidden state with the teacher’s penultimate scalar invariants. These variations let us ask whether the transfer is driven by the distillation recipe or by the architecture of the student.

3 Experimental Setup

3.1 Dataset and Metrics

We use the standard TopTagging benchmark of Kasieczka et al. [4], with 1.21M training, 403k validation and 404k test jets. Each jet is a variable-size set of constituent four-momenta, zero-padded to a fixed maximum of 200 particles. As in the equivariant-tagger literature, we report test AUC and background rejection at fixed signal efficiencies ($\epsilon_S \in \{0.3, 0.5, 0.8\}$). This benchmark is useful for controlled architecture comparisons, but its i.i.d. test split is already close to saturated; small AUC differences are therefore not by themselves evidence of symmetry transfer.

3.2 Teacher

We use an L-GATr checkpoint trained on TopTagging as the teacher. The teacher input follows the validated L-GATr pipeline: seven jet-tagging scalar features, a global-token flag, particle four-momenta, and the reference/spurion tokens used by the equivariant model. The teacher is frozen throughout, and its logits are cached before student training. Headline comparisons use one teacher checkpoint; seed variability is quantified separately in Section 4.4. Implementation details for loading the checkpoint are given in Sections A and B.

3.3 Student Models

We compare non-equivariant student architectures with different amounts of built-in structure. None use equivariant operations or geometric-algebra primitives.

- **Flat MLP, engineered features.** The dense student of Liu et al. [5]. Per-particle features ($p_T^{\text{rel}}, \Delta\eta, \Delta\phi, E^{\text{rel}}$) are computed against the jet four-momentum, padded/truncated to 128 particles, ordered by p_T , and flattened into a single input vector. This model tests how far distillation can go without explicit inter-particle structure.
- **DeepSets PFN.** A permutation-invariant per-particle network over the 7-dimensional feature set ($\log p_T, \log E, \log p_T^{\text{rel}}, \log E^{\text{rel}}, \Delta\phi, \Delta\eta, \Delta R$), followed by masked sum, mean and max pooling.
- **Plain transformer (no pair bias).** A standard pre-LN encoder stack with multi-head self-attention over the 7-dim per-particle features, masked-mean pooling, and a linear classification head. We instantiate three configurations to disentangle raw capacity from architectural prior: a *tiny* variant ($d_{\text{model}}=32$, 2 blocks; $\sim 50\text{k}$ params), a *base* variant ($d_{\text{model}}=64$, 4 blocks, 4 heads, FFN width 256, dropout 0.1; $\sim 205\text{k}$ params), and a *larger* variant ($d_{\text{model}}=96$, 6 blocks, FFN width 384; $\sim 600\text{k}$ params). The base configuration is the reference for the supervision-side ablations of Section 4.2; the larger configuration probes whether plain capacity alone can close the iid gap to the teacher.

- **ParT-style transformer (pair-bias).** The architecture above (4 blocks, $d_{\text{model}}=64$, 4 heads, $\sim 205\text{k}$ params) augmented with a learned pairwise-feature attention bias [7]: a small MLP maps pre-computed pair features ($\Delta R_{ij}, \log k_{T,ij}, \log m_{ij}^2$) into a per-head additive logit injected into each attention block. *This is the only student class with a nontrivial two-particle inductive prior; it is the headline configuration* (Sections 4.1 and 4.6).

3.4 Distillation loss

The KD objective follows the convex-combination convention of Hinton et al. [3] and Liu et al. [5],

$$\mathcal{L} = \alpha T^2 \text{MSE}(z_S/T, z_T/T) + (1-\alpha) \text{BCE}(z_S, y), \quad (1)$$

where $z_S, z_T \in \mathbb{R}$ are scalar pre-sigmoid logits, $\alpha \in [0, 1]$ weights the KD term, and $T > 0$ is the temperature. ?? shows the first term simplifies to $\text{MSE}(z_S, z_T)$ identically in T ; hence temperature is mathematically degenerate in this loss form. We adopt logit-MSE with $\alpha=0.5$ as the default in the architecture sweep of Section 4.1 and sweep α along with T in Section 4.2. As a controlled alternative we also report Hinton-style softmax- T KL on the sigmoid-induced two-class distribution, where T is genuinely a regularizer (Section 4.2 and ??).

3.5 Supervision signals

Hint loss. We test whether intermediate teacher features provide useful supervision beyond the final logit, following Romero et al. [8]. The hint target is a 48-dimensional per-jet invariant representation from the final L-GATr block: the scalar component of each of the 16 multivector channels plus all 32 scalar-stream channels at the global token. A small projection head maps the student’s pooled representation to this target via MSE, weighted by β and added to Equation (1). This isolates the supervision-richness lever from the architecture lever.

Pairwise input features. For DeepSets, we additionally provide four hand-engineered per-particle aggregates summarizing each particle’s relationship with the rest of the jet: $\Delta R_{\min}$ to the nearest neighbor, $\log k_{T,\min}$ over neighbors, count of neighbors with $\Delta R < 0.3$, and ΔR to the hardest particle in the jet. These are non-equivariant scalars the student can use without architectural change.

3.6 Training

All students are trained with the same optimizer, learning-rate schedule and batching convention. Teacher logits, and where used the penultimate invariant targets, are pre-computed on the train and validation splits so that student training does not require teacher forwards. The default schedule is 30 epochs; the headline ParT-style student and the larger plain-transformer baseline use 50 epochs for the capacity comparison in Section 4.5. Full hyperparameters and boost-evaluation details are given in Section B.

4 Experiments

4.1 Architecture is the binding constraint

KD lifts every architecture, with diminishing returns up the inductive-bias staircase. With the corrected teacher pipeline, KD lifts the flat MLP from a scratch baseline of 0.9501 to 0.9743 (+0.0242 AUC; $3.0\times \text{rej}@_{\varepsilon_S=0.5}$ multiplier from 45 to 137), the DeepSets PFN to 0.9833, the small transformer to 0.9852, and the ParT-style pair-bias student to 0.9869. The lift narrows monotonically: KD provides much more for an architecture that cannot represent inter-particle structure than for one that already can. On iid AUC the ParT-style KD student matches the

Table 1: Distilled students vs. the L-GATr teacher on top tagging. Bolded row is the headline configuration. For the plain (no-pair-bias) transformer, we report only the best-performing of three swept configurations; the full sweep is given in Section 4.5.

MODEL	PARAMS	ACC	AUC	REJ@0.3	REJ@0.5	REJ@0.8
L-GATr (TEACHER, EQUIVARIANT)	$\sim 1.0\text{M}$	—	0.9870	2433	587	67
MLP	528k	0.9174	0.9743	464	137	26
DEEPSETS (PAIRWISE FEATURES)	167k	0.9357	0.9837	1195	311	47
PLAIN TRANSFORMER	$\sim 600\text{k}$	0.9396	0.9858	1787	440	58
PAR-T-STYLE TRANSFORMER (PAIR-BIAS)	205k	0.9422	0.9869	2433	567	67

Table 2: Distilled vs. non-distilled training at matched architecture and parameter count. KD lifts both the weakest student class (flat MLP) and the strongest (ParT-style) over their hard-label-only counterparts; the iid AUC gap at the ParT architecture is small but, as Section 4.6 shows, hides a much larger boost-robustness gap that grows by $30\times$ under Lorentz boosts.

CONFIG.	PARAMS	ACC	AUC	REJ@0.5
<i>MLP</i>				
SCRATCH	528k	0.8875	0.9501	45
DISTILLED	528k	0.9174	0.9743	137
<i>PAR-T-STYLE TRANSFORMER (PAIR-BIAS)</i>				
SCRATCH	205k	0.9401	0.9857	388
DISTILLED	205k	0.9422	0.9869	567

teacher within seed noise (Section 4.4): $\Delta\text{AUC} = 0.0001$, $\text{rej}@_{\varepsilon_S} \in \{0.3, 0.8\}$ exact, 93% of the $\text{rej}@_{\varepsilon_S=0.5}$ gap closed.

Capacity is not the lever. Within the KD-trained block, the ordering is by architectural inductive bias rather than parameter count. The flat MLP at 528k params (the largest student in our family) trails the 50k tiny transformer by $+0.009$ AUC and the 150k DeepSets PFN by $+0.009$ AUC: an order of magnitude more parameters, half the rejection. The 205k pair-bias student outperforms a 600k plain-attention student by $+0.0011$ AUC and $+127$ $\text{rej}@_{\varepsilon_S=0.5}$ (Section 4.5, Figure 1).

The KD-vs-scratch iid gap looks small but is the visible tip of a much larger one. At the ParT architecture, KD beats scratch by 0.0012 AUC on iid data — close to the seed-noise floor and easy to dismiss. Section 4.6 shows the same gap widens by $30\times$ under Lorentz boosts of the test inputs and is the actual signature of inductive-bias transfer.

4.2 Supervision-side levers do not move AUC

Across both ends of the inductive-bias spectrum, supervision-side levers are inert. The 12-cell DeepSets sweep is flat to within ± 0.0001 AUC. The T -flatness is structural (??); the α - and hint-flatness are empirical. To isolate the soft-target-form lever from the T -degeneracy of logit-MSE, we re-trained the flat MLP and the ParT student under Hinton-style softmax- T KL where temperature genuinely modulates the gradient. On the MLP, $\Delta\text{AUC} = 0.0005$ (0.9738 vs. 0.9743); on the ParT student, $\Delta\text{AUC} = 0.0001$ (0.9870 vs. 0.9869). Both are inside the

Table 3: Intermediate-feature hint distillation vs. logit-only distillation, holding the student architecture fixed. The hint loss converges (training MSE drops by $\sim 6\times$ across epochs) but does not transmit additional classification utility, even with a structurally homologous transformer student.

CONFIGURATION	ACC	AUC	REJ@0.3	REJ@0.5
<i>DEEPSETS PFN ($\sim 150K$ PARAMS)</i>				
LOGIT ONLY	0.9349	0.9833	1188	290
LOGIT + HINT ($\beta=0.5$)	0.9348	0.9833	1141	289
<i>PLAIN TRANSFORMER, BASE ($d=64$, 4 BLOCKS; $\sim 205K$ PARAMS)</i>				
LOGIT ONLY	0.9369	0.9845	1541	368
LOGIT + HINT ($\beta=0.5$)	0.9368	0.9845	1453	352

seed-noise floor of Section 4.4. The supervision channel does not appear to be the bottleneck on this benchmark, regardless of whether T is mathematically meaningful in the chosen loss form.

4.3 Hint loss does not rescue under structural homology

A natural hypothesis for the DeepSets hint-loss null is that masked sum/mean/max pooling destroys the per-particle structure the teacher’s penultimate invariants encode, so a regression target onto a permutation-collapsed bottleneck is uninformative. We tested this on the small transformer student, whose per-particle attention preserves structural homology with the teacher. The result: AUC 0.9845 with the hint vs. 0.9845 without it; $\text{rej@}\varepsilon_S=0.5$ of 352 vs. 368. The hint loss converges (training MSE drops by $\sim 6\times$ across epochs) but does not transmit additional classification utility regardless of student architecture. We interpret the penultimate invariants as a *downstream readout* of the teacher’s geometric reasoning, not its substrate; matching them by regression does not transmit the upstream computation that produced them.

4.4 Seed-noise floor

We ran the small transformer with seeds 42 and 43, holding everything else fixed. Test AUC: 0.984489 vs. 0.984487 ($\Delta = 3 \times 10^{-6}$). Rejection at $\varepsilon_S = 0.5$: 367.8 vs. 372.5 ($\Delta = 5$). Best validation AUC agrees to four decimal places. We adopt ± 0.0001 AUC and ± 5 in $\text{rej@}\varepsilon_S=0.5$ as the inter-seed noise floor on this benchmark and architecture; deltas in Section 4.1 sit safely outside it, while every cell of ?? sits inside.

4.5 Right architecture beats raw capacity

The parameter-efficiency observation is that the inductive prior carried by pair-bias attention [7] is worth $\sim 3\times$ the parameter count of plain attention on this task. The pair-bias term is purely structural — it adds $\sim 2k$ parameters in the bias MLP, negligible on a 205k base — yet outperforms the 600k plain-attention model. Figure 1 visualizes this across all student configurations on a common parameter axis: pair-bias is the only point clearly above the plain-capacity scaling trend, while the flat MLP at 528k params sits well below smaller, more-structured students.

4.6 Boost robustness: KD transfers $\sim 40\%$ of the teacher’s partial Lorentz inductive bias

The iid-AUC gap between KD and scratch students at the ParT-style architecture is small (0.0012), in line with the “saturated benchmark” regime of top tagging on the standard split. The stronger and more diagnostic claim, following Liu et al. [5], is that KD transfers *behavioral*

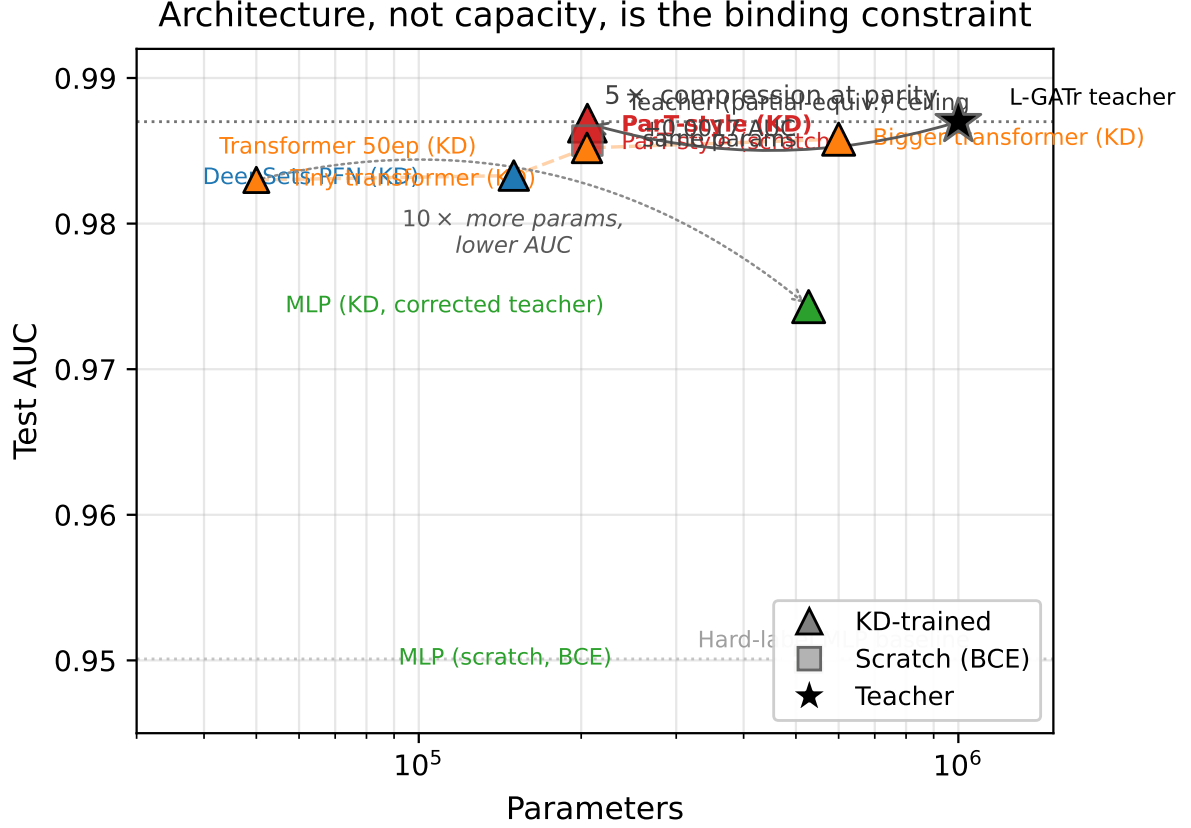


Figure 1: Parameter efficiency vs. test AUC across all KD-trained students, scratch baselines, and the L-GATr teacher. *Architecture, not capacity, is the lever*. The flat MLP at 528k params (largest student in our family) underperforms the 50k tiny transformer; the 205k ParT-style student matches the 1M-param teacher and outperforms a 600k plain-attention transformer at $\sim 3\times$ fewer parameters. Triangles: KD-trained students; squares: matched-architecture scratch baselines; star: teacher.

Lorentz robustness measurable through how the student degrades when the test inputs are subjected to Lorentz boosts the student has never seen. We x -boost the dense test-set 4-momenta with $\beta \in \{0, 0.05, \dots, 0.95\}$, recompute the (Lorentz-non-invariant) student input features on the boosted inputs, and re-evaluate. The procedure mirrors Liu et al. [5].

The teacher is only partially equivariant. L-GATr carries equivariance through its multivector paths, but the scalar input channels we feed it ($\log p_T$, $\log E$, $\Delta\eta$, etc.) are explicitly Lorentz-non-invariant. Empirically, the teacher’s AUC drops from 0.9870 at $\beta = 0$ to 0.9611 at $\beta = 0.9$ (Table 4). The student therefore has only a partial inductive bias to inherit, not full equivariance.

Three findings.

1. **The KD-vs-scratch AUC gap widens by $30\times$ from $\beta=0$ to $\beta=0.9$** — from $+0.0012$ at iid to $+0.0408$ in the high-boost regime. The iid AUC alone almost completely hides the inductive-bias transfer; only the boost-eval reveals it. This replicates the qualitative claim of Liu et al. [5] in our setting (different teacher class, different student class, different input feature pipeline) and quantifies it.
2. **The transfer fraction is stable at 41.2% over $\beta \in [0.5, 0.85]$.** The student picks up

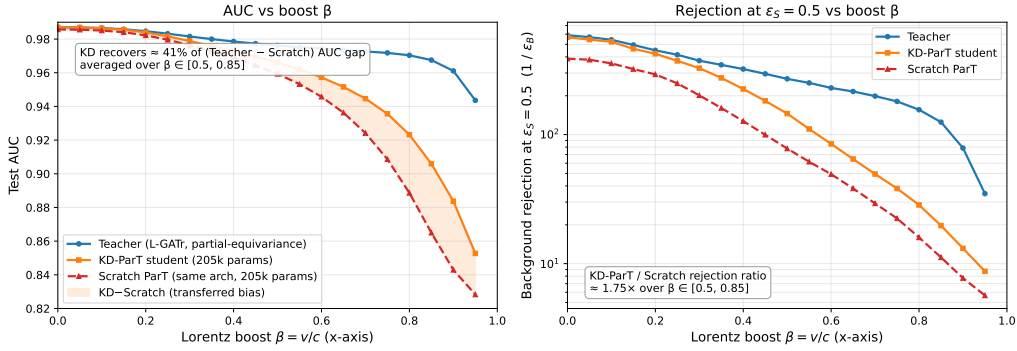


Figure 2: Boost robustness on the top-tagging test set. *Left*: AUC vs. boost magnitude β . The KD-trained ParT-style student (orange) sits between the teacher’s curve (green, partial-equivariance ceiling) and the same-architecture scratch baseline (blue) at every β , never below scratch and never above teacher. *Right*: rejection at $\epsilon_S = 0.5$, log scale. The KD-vs-scratch ratio averages $1.75\times$ over $\beta \in [0.5, 0.85]$ and stays above $1.4\times$ across the full range.

Table 4: AUC and rejection at selected boost magnitudes. The rightmost two columns isolate the KD-vs-scratch effect. Transfer fraction is defined as $(\text{KD} - \text{Scratch})/(\text{Teacher} - \text{Scratch})$; mean over $\beta \in [0.5, 0.85]$ is 41.2%. The $\text{rej@}\epsilon_S=0.5$ KD/Scratch ratio is the deployment-relevant figure — LHC trigger systems optimize for fixed signal efficiency.

β	TEACHER AUC	KD AUC	SCRATCH AUC	KD-SCRATCH	TRANSFER	REJ@0.5 RATIO
0.0	0.9870	0.9869	0.9857	+0.0012	—	$1.46\times$
0.4	0.9786	0.9729	0.9689	+0.0040	41%	$1.78\times$
0.5	0.9762	0.9660	0.9593	+0.0067	40%	$1.86\times$
0.7	0.9727	0.9446	0.9241	+0.0205	42%	$1.69\times$
0.8	0.9704	0.9232	0.8886	+0.0346	42%	$1.75\times$
0.9	0.9611	0.8837	0.8429	+0.0408	—	$1.63\times$

roughly two-fifths of the way from the same-architecture scratch baseline to the (only partially-equivariant) teacher in boost-AUC, robustly across the regime where the teacher itself shows nontrivial structure to transfer.

3. **The $\text{rej@}\epsilon_S=0.5$ multiplier is consistently $1.7\text{--}1.9\times$** across the boost range, with mean $1.75\times$ over $\beta \in [0.5, 0.85]$. Even at $\beta = 0$ the KD multiplier is $1.46\times$; it grows with boost magnitude. The deployment-relevant claim is therefore: KD-distilled ParT students are $\sim 75\%$ more boost-robust than the identical architecture trained on hard labels alone.

The teacher curve in Figure 2 (green) is the *partial-equivariance ceiling*, not flat. The KD student (orange) lies strictly between teacher and scratch at every β , always closer to teacher than to scratch — the cleanest visual demonstration of inductive-bias transfer the experiment could produce.

5 Discussion and Conclusion

We have shown that a compact ParT-style student can be distilled from an L-GATr teacher to match the teacher’s iid TopTagging AUC within the measured seed-noise floor, while using about five times fewer parameters. This result is not simply a statement about making the

loss more elaborate. Across the student family, the decisive factor is whether the architecture can represent particle-pair structure. Pair-bias attention supplies that relational prior directly; increasing plain-attention capacity or adding supervision-side refinements does not give the same gain in these experiments.

Architecture is the main lever. The supervision ablations are deliberately broad: temperature, hard/soft mixing, soft-target form, pairwise input aggregates and intermediate-feature hints all test plausible ways to make the distillation recipe carry more information. Their small effect suggests that the teacher logit already contains enough task-level guidance once the student has the right inductive structure. The larger failure mode is architectural mismatch: a student that cannot express the relevant two-particle relations cannot recover them from a more carefully tuned KD objective alone.

Boost evaluation reveals the transferred behavior. On the iid split, the distilled and scratch ParT-style students are separated by a small AUC gap. Under Lorentz x -boosts of the test inputs, that gap widens substantially: the distilled student remains closer to the teacher reference curve than the same architecture trained only on hard labels, recovering about two-fifths of the teacher–scratch AUC gap over the main boost range. We therefore interpret the boost sweep as evidence of behavioral robustness transfer, not as evidence that the non-equivariant student has acquired L-GATr’s exact architectural symmetry.

Limitations. The study is limited to TopTagging, one L-GATr teacher checkpoint for the headline comparisons and an x -direction boost protocol. Student seed variability is measured more carefully than teacher variability, and we do not benchmark inference latency. It remains open whether the same architecture-dominant pattern holds for less saturated datasets, quark/gluon discrimination, multi-class jet classification or generative tasks.

Future directions. The natural next tests are to repeat the comparison in regimes where the teacher is not already near saturation, to study whether more Lorentz-invariant student inputs reduce boost sensitivity, and to determine how far the pair-bias student can be compressed before the boost-robustness gains disappear.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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Appendix

A Teacher-loading methodology footnote

The L-GATr top-tagging teacher exposes an input-side config with `in_s_channels=8`: seven jet-tagging features ($\log p_T$, $\log E$, $\log p_T^{\text{rel}}$, $\log E^{\text{rel}}$, $\Delta\phi$, $\Delta\eta$, ΔR) plus a one-bit global-token flag. The wrapper additionally expects three spurions (two lightlike beams and a time reference), $/20$ momentum scaling, and a small mass-regularization step. Instantiating the teacher with `in_s_channels=1` and loading the published checkpoint via `strict=False` (the natural “minimal interface” guess we initially used) silently masks the input-projection mismatch: the network forward-passes without error but produces meaningless soft targets. Our earliest MLP-KD experiments reported AUC ~ 0.78 in this regime — below the same architecture’s hard-label-only baseline of 0.9501. After the bug was identified, the same MLP architecture trained against the corrected teacher pipeline reaches AUC 0.9743 (Table 1). The methodology cost of the single integer config flag was therefore $\Delta\text{AUC} \approx 0.185$, which we record here as a warning to subsequent users of the released checkpoint.

B Hyperparameters and training details

All students train with AdamW (lr 3×10^{-4} , weight decay 10^{-4}), cosine schedule over the full epoch budget, batch size 512, gradient clipping at unit norm, and mixed-precision autocast (fp16) with a `GradScaler`. The default schedule is 30 epochs; the headline ParT run and the bigger-transformer baseline use 50 (see Section 4.5). Teacher logits, and where used the 48-dim penultimate invariants, are pre-computed once over the full train and validation splits and cached to disk, so per-epoch student steps incur no teacher-forward cost. The hint-projection head is a two-layer GELU MLP with width 128 mapping the student’s pooled (DeepSets) or CLS-token (transformer) representation to the 48-dim invariant target. Pairwise-input features for the DeepSets variant are $\Delta R_{\min}$, $\log k_{T,\min}$, count of neighbors with $\Delta R < 0.3$, and ΔR to the hardest particle in the jet — four scalar aggregates per particle, computed once per batch on GPU. Boost-evaluation follows Liu et al. [5]: dense-tensor x -boost of the test 4-momenta with $\beta \in \text{linspace}(0, 1, 20, \text{endpoint}=\text{False})$, recompute student input features and pair features on the boosted inputs, single forward pass per β per checkpoint.